

IPHY 6010

Seminar

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	AI Engineer (3.9 / 100)	1.2 / 100
SPECIALIZATION On-Campus Elective		

1

CURRENT AVG SCORE
across 6 job roles

79

POTENTIAL SCORE
with recommended additions

+84

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Presents special topics in integrative physiology.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	3.9	73.0	+69.1	HIGH
Data Scientist	3.4	83.8	+80.4	HIGH
Applied Scientist	0.0	70.9	+70.9	HIGH
ML Engineer	0.0	82.3	+82.3	HIGH
MLOps Engineer	0.0	81.9	+81.9	HIGH
Machine Learning Engineer	0.0	83.6	+83.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Model Deployment & Serving 2 of 6 roles

production

Business Communication 1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 MEDIUM Consider Model Deployment & Serving content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

3 LOW Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

AI Engineer

3.9 → 73.0 (+69.1 pts)

data engineering models learning build form software systems

Data Scientist

3.4 → 83.8 (+80.4 pts)

data product characteristics presentation evolution insights different build

Applied Scientist

0.0 → 70.9 (+70.9 pts)

lab research reasoning models data learning systems engineering

ML Engineer

0.0 → 82.3 (+82.3 pts)

data learning systems models production engineering starting build

MLOps Engineer

0.0 → 81.9 (+81.9 pts)

data production systems learning engineering tools models development

Machine Learning Engineer

0.0 → 83.6 (+83.6 pts)

data engineering learning tools solutions time design development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.